IE-Problem set 2

1. Consider a principal-agent problem with moral hazard, a continuum of actions $a \in [0, \overline{a}]$ and outcomes $x \in [\underline{x}, 0]$, $\underline{x} < 0$. The principal is risk neutral, and the agent is strictly risk averse with utility function $u(w) - c(a)$ for income-action pairs (w, a), with u unbounded below, $u' > 0$, $u'' < 0$, $c(0) = 0$, $c' > 0$, and $c'' \geq 0$. Suppose $f(x|a)$ is as follows: $f(0|a) = 1 - p(a)$, $0 < p(a) < 1$ for all a, and $f(x|a) = p(a)g(x)$ for $x < 0$ and all a, where $g(x) > 0$ for all $x \in [\underline{x}, 0)$ and $\int_{\underline{x}}^{0} g(x)dx = 1$, and $p''(a) > 0 > p'(a)$ for all a. The agent's reservation utility is $\bar{u}$.

- **(a)** Set up the principal's (expected profit-maximizing) contracting problem. Is the first-order approach valid? Explain.
- **(b)** Assume the first-order approach and obtain, for each $a \in [0, \overline{a}]$, the cost $C(a)$ for the principal and the cost-minimizing contract $w(\cdot)$ (expressed in money) or $v(\cdot)$ (expressed in utils) associated with each a.
- **(c)** Using the cost function C derived in (b), set up the problem the principal would solve to find the action a that maximizes expected profits.
- **(d)** Provide an insurance contract interpretation of this problem.

2. Consider a principal-agent problem with moral hazard, a continuum of effort levels $a \geq 0$ and two output levels $x \in \{x_\ell, x_h\}$, $0 < x_\ell < x_h$. The principal is risk neutral, and the agent is strictly risk averse with utility function that is additively separable in income and effort, given by $\log(\xi + w) - \psi(a)$ for wage-effort pairs (w, a), with $\psi(0) = \psi'(0) = 0$, $\psi' > 0$ for $a > 0$, and $\psi'' \geq 0$, and where $\xi > 0$ is the agent's initial wealth (which is a parameter of the problem and is publicly observable). The probability of x_h when the agent exerts effort a is $p(a)$, and thus the probability of x_ℓ is $1 - p(a)$, where $p(0) \in (0, 1)$, $p' > 0$ and $p'' < 0$. The agent's reservation utility is $\log(\xi + \bar{w})$, where $\bar{w}$ is the wage the agent can obtain with certainty in her outside option (also publicly observable), which we assume is large enough so that the nonnegativity constraint (due to log utility) on initial wealth plus wage for each output realization can be ignored.

- **(a)** Set up the principal's contracting problem, and explain the two steps you would follow to solve the problem. Is the first-order approach valid? Justify your answer.
- **(b)** For each effort level $a \geq 0$, obtain the compensation scheme (in terms of a utility level for each realization of x) that minimizes the principal's expected cost of inducing the agent to exert that effort level. Use any theoretical result you know that may allow you to simplify the algebra. Explain the properties of the cost-minimizing compensation scheme.
- **(c)** Show that the principal weakly prefers a poorer agent (that is, one with a lower ξ).